

Model Lab

Every prior, same slate — 2026-09-16

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

35,673

Held-Out Scores

Fitted on plate appearances before 2026-08-11, scored on the 35,673 that came after (actual HR rate 3.03%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5670	1.36×	0.02931	0.13495	0.00416
KNN comps Prior is the pooled rate of the nearest comparables.	0.5857	1.52×	0.02930	0.13460	0.00527
SVR Support-vector regression on the same feature matrix.	0.5734	1.28×	0.02941	0.13660	0.00900
Random forest Forest regression; a learned similarity metric.	0.5763	1.35×	0.02937	0.13597	0.00807
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5857	1.52×	0.02930	0.13460	0.00527
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5852	1.50×	0.02930	0.13464	0.00589
Linear regression Ordinary least squares on the same features.	0.5804	1.28×	0.02934	0.13550	0.00728
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5831	1.52×	0.02934	0.13516	0.00765
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5819	1.33×	0.02934	0.13507	0.00734
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5735	1.28×	0.02942	0.13645	0.00980

Blend weights (forest variant): KNN 0.95, SVR 0.00, RF 0.05

Shrinkage K per prior: league 246, knn 246, svr 246, rf 246, core 246, ensemble 246, linear 246, logistic 246, nn 246, xgb 246 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,822 PA, actual HR rate 2.46%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5031	1.06×	0.02402	0.11602	0.01052
KNN comps Prior is the pooled rate of the nearest comparables.	0.5542	0.96×	0.02411	0.11766	0.01370
SVR Support-vector regression on the same feature matrix.	0.5289	0.96×	0.02466	0.12930	0.02652
Random forest Forest regression; a learned similarity metric.	0.5377	0.85×	0.02449	0.12566	0.02325
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5542	0.96×	0.02411	0.11766	0.01370
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5516	0.96×	0.02412	0.11791	0.01327
Linear regression Ordinary least squares on the same features.	0.5520	0.85×	0.02434	0.12390	0.01968
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5615	1.38×	0.02423	0.11970	0.01893
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5596	0.96×	0.02422	0.11935	0.01816
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5331	0.85×	0.02486	0.12853	0.02981

Slate Agreement

Flat league · 5/6 shared 1/6 hit

Kyle Schwarber
Junior Caminero
Ronald Acuña Jr.
Bryce Harper
Matt Olson
Pete Alonso ✓

SVR · 4/6 shared 0/6 hit

Kyle Schwarber
Junior Caminero
Shea Langeliers
Griffin Conine
Matt Olson
James Wood

Core (KNN+SVR) · 6/6 shared 1/6 hit

Kyle Schwarber
Matt Olson
Pete Alonso ✓
Junior Caminero
Shea Langeliers
Bryce Harper

Linear regression · 4/6 shared 1/6 hit

Kyle Schwarber
Emmanuel Rodriguez
Junior Caminero
Matt Olson
Griffin Conine
Pete Alonso ✓

Neural net (torch) · 4/6 shared 0/6 hit

KNN comps · 6/6 shared 1/6 hit

18.4%	Kyle Schwarber	21.5%
16.6%	Matt Olson	18.0%
15.6%	Pete Alonso ✓	17.6%
15.5%	Junior Caminero	17.6%
15.3%	Shea Langeliers	17.2%
15.3%	Bryce Harper	17.1%

Random forest · 4/6 shared 0/6 hit

23.1%	Kyle Schwarber	22.1%
19.8%	Matt Olson	18.5%
19.4%	Junior Caminero	18.4%
19.3%	James Wood	18.1%
19.2%	Shea Langeliers	17.9%
18.3%	Griffin Conine	17.9%

Forest blend · 5/6 shared 1/6 hit

21.5%	Kyle Schwarber	21.5%
18.0%	Matt Olson	18.0%
17.6%	Junior Caminero	17.6%
17.6%	Pete Alonso ✓	17.6%
17.2%	Shea Langeliers	17.2%
17.1%	James Wood	17.2%

Logistic regression · 3/6 shared 0/6 hit

22.2%	Kyle Schwarber	25.0%
20.0%	Emmanuel Rodriguez	23.2%
17.9%	Matt Olson	19.4%
17.9%	Shea Langeliers	19.3%
17.8%	Griffin Conine	18.6%
17.6%	Coby Mayo	18.6%

James Wood

17.8%

Pete Alonso ✓

17.9%

Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Kyle Stowers Miami Marlins · vs RHP · 19 HR / 482 PA	14.2%	17.1%	17.2%	16.6%	17.1%	17.1%	16.0%	17.1%	16.8%
9. Griffin Conine Miami Marlins · vs RHP · 16 HR / 286 PA	14.4%	17.1%	19.3%	17.9%	17.1%	17.1%	17.8%	18.6%	18.4%
10. Ronald Acuña Jr. Atlanta Braves · vs RHP · 18 HR / 436 PA	15.6%	16.8%	17.0%	16.6%	16.8%	16.8%	16.5%	16.6%	16.8%
11. Heriberto Hernández Miami Marlins · vs RHP · 24 HR / 460 PA	14.6%	16.8%	17.2%	17.1%	16.8%	16.9%	16.9%	17.4%	17.4%
12. Riley Greene slate Detroit Tigers · vs RHP · 20 HR / 549 PA	14.2%	16.7%	16.7%	16.7%	16.7%	16.7%	16.1%	17.0%	16.8%
13. Abimelec Ortiz Washington Nationals · vs LHP · 5 HR / 121 PA	12.7%	16.7%	15.4%	16.1%	16.7%	16.6%	16.3%	15.7%	16.7%
14. Coby Mayo Baltimore Orioles · vs RHP · 24 HR / 416 PA	13.8%	16.5%	17.8%	17.1%	16.5%	16.5%	16.8%	18.6%	17.3%
15. Cal Raleigh Seattle Mariners · vs RHP · 22 HR / 490 PA	13.5%	16.2%	15.0%	15.7%	16.2%	16.2%	16.6%	17.7%	16.4%
16. Pete Crow-Armstrong slate Chicago Cubs · vs LHP · 42 HR / 682 PA	14.3%	16.1%	17.1%	16.6%	16.1%	16.1%	16.1%	17.4%	16.8%
17. Spencer Torkelson Detroit Tigers · vs RHP · 24 HR / 571 PA	13.7%	16.0%	15.3%	15.2%	16.0%	16.0%	15.0%	15.2%	15.2%
18. Aaron Judge New York Yankees · vs LHP · 18 HR / 283 PA	12.0%	16.0%	17.3%	17.0%	16.0%	16.1%	17.1%	18.3%	17.1%
19. Zack Gelof Athletics · vs LHP · 18 HR / 389 PA	14.1%	16.0%	16.3%	16.1%	16.0%	16.0%	16.1%	16.0%	16.2%
20. Oneil Cruz Pittsburgh Pirates · vs RHP · 18 HR / 384 PA	13.4%	15.9%	16.7%	15.7%	15.9%	15.9%	15.4%	15.4%	15.8%
21. Juan Soto New York Mets · vs RHP · 25 HR / 431 PA	13.0%	15.8%	16.6%	16.2%	15.8%	15.8%	15.9%	16.1%	16.2%
22. Kazuma Okamoto	14.0%	15.8%	16.9%	16.1%	15.8%	15.8%	15.8%	16.8%	16.4%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Dillon Dingler Detroit Tigers · vs RHP · 27 HR / 575 PA	13.4%	14.6%	15.3%	14.3%	14.6%	14.6%	14.5%	14.9%	14.8%
38. Victor Mesa Jr. Tampa Bay Rays · vs RHP · 13 HR / 226 PA	13.7%	14.5%	14.8%	14.5%	14.5%	14.5%	14.8%	14.5%	15.0%
39. Jeremy Peña Houston Astros · vs RHP · 19 HR / 434 PA	14.2%	14.4%	14.8%	14.6%	14.4%	14.4%	14.3%	14.3%	14.2%
40. Dominic Canzone Seattle Mariners · vs RHP · 22 HR / 487 PA	12.6%	14.4%	14.7%	15.1%	14.4%	14.4%	14.4%	15.0%	14.9%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.